

BIKE RENTAL DEMAND FORECASTING SYSTEM BASED ON WEATHER CONDITIONS, TIME, AND SEASONAL TRENDS

Problem Statement:

Bike-sharing services experience significant fluctuations in rental demand due to changing weather conditions, time of day, holidays, and seasonal variations. Traditional demand estimation methods often rely on historical averages, making it difficult to accurately predict future demand and optimize bike availability. Inefficient demand forecasting can lead to bike shortages, excess inventory, and reduced customer satisfaction.

The proposed solution is a **Bike Rental Demand Forecasting System** that predicts future rental demand using machine learning and historical usage data. The system provides a centralized platform where rental trends and environmental factors are analyzed through a secure and user-friendly interface.

By integrating regression algorithms and predictive analytics, the system evaluates weather conditions, temperature, humidity, wind speed, season, holidays, and time-based patterns to forecast bike rental demand accurately. Interactive dashboards display demand trends, prediction reports, and resource utilization statistics, helping operators optimize bike distribution and improve operational efficiency.

The system is secure, scalable, and suitable for bike-sharing companies, smart city initiatives, and transportation service providers.

Ultimately, this project delivers a modern and intelligent demand forecasting solution that improves prediction accuracy, optimizes bike availability, reduces operational costs, and enhances customer satisfaction.